

Detailed Marking Report — Mathematics

exam-10-ch1-2-4-5-7-10 | Units 1, 2, 4, 5, 7, 10 | Federal Board Annual Examination 2026
Candidate: Seemab | Attempt date: 26 June 2026 | Marked: 26 June 2026 (corrected)

61/65
TOTAL SCORE

93.8%
PERCENTAGE

A+
GRADE

12/12
SECTION A (MCQ)

32/33
SECTION B

17/20
SECTION C

All three sections present and clearly readable: Section A (12 MCQs, page 1), Section C (pages 1-4), Section B (pages 5-10). Seemab answered Section C before Section B, which is allowed. All 11 required Section B questions attempted; all 4 Section C questions attempted.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

Seemab's answers read from handwritten list on page 1. Q4 showed "A" struck through and corrected to "B"; B is the key answer — full credit.

#	Question (short)	Correct	Seemab	Result	Marks
1	Which number is irrational?	A ($\sqrt{7}$)	A	✓	1
2	Simplest form of $\sqrt{50}$	B ($5\sqrt{2}$)	B	✓	1
3	Value of $\log_{10} 1000$	C (3)	C	✓	1
4	$\log(mn)$ equals?	B ($\log m + \log n$)	B	✓	1
5	Factors of $x^2 - 7x + 12$	A ($(x-3)(x-4)$)	A	✓	1
6	Factorize $a^3 + 8$	D ($(a+2)(a^2-2a+4)$)	D	✓	1
7	Solve $2x + 3 = 11$	C ($x = 4$)	C	✓	1
8	Solution set of $ x = 5$	D ($\{-5, 5\}$)	D	✓	1
9	Distance (0,0) to (3,4)	B (5)	B	✓	1
10	Midpoint of (2,4) and (6,10)	D (4, 7)	D	✓	1
11	Centre to point on circle	A (Radius)	A	✓	1
12	Measurements for unique triangle	C (3)	C	✓	1
SECTION A TOTAL					12/12

Perfect 12/12 on MCQs. This extends the current consecutive-perfect MCQ streak to 7 exams (exams 14-20). Math MCQ: 59/60 across all 5 Math exams.

SECTION B — Short Questions (11 × 3 = 33 Marks)

Seemab answered the OR variant for Q2(i), Q2(ii), and Q2(iii). She answered the main question for Q2(iv) through Q2(xi).

Q.2(i) OR — Rational or irrational: (a) $\sqrt{36}$ (b) $\sqrt{10}$ (c) -0.7

3/3

(a) $\sqrt{36}$: Wrote $\sqrt{36} = 6$, classified as rational and perfect square. ✓ Correct. (1/1)

(b) $\sqrt{10}$: Classified as irrational, not a perfect square. ✓ Correct. (1/1)

(c) -0.7 : Classified as rational, converted to $-7/10$. ✓ Correct. (1/1)

Q.2(ii) OR — Scientific notation: (a) 0.00046 (b) 92,000,000

3/3

(a) 0.00046: Wrote 4.6×10^{-4} . ✓ Correct. (1.5/1.5)

(b) 92,000,000: Wrote 9.2×10^7 . On close reading of the scan the superscript is clearly 7. 9.2×10^7 is correct. ✓ Full credit. (1.5/1.5)

Q.2(iii) OR — Expand $\log(a^2b / c)$

3/3

Seemab wrote: $\log a^2b/c = \log a^2 + \log b - \log c = 2 \log a + \log b - \log c$.

✓ All three steps (quotient law, product law, power law) correctly applied. Full credit. (3/3)

Q.2(iv) main — Given $\log 2 = 0.3010$, $\log 3 = 0.4771$: (a) $\log 6$ (b) $\log 12$

3/3

(a) $\log 6$: $\log(2 \times 3) = \log 2 + \log 3 = 0.3010 + 0.4771 = 0.7781$. ✓ Correct. (1.5/1.5)

(b) $\log 12$: $\log(2^2 \times 3) = 2 \log 2 + \log 3 = 0.602 + 0.4771 = 1.0791$. ✓ Correct. (1.5/1.5)

Q.2(v) main — Factorize: (a) $x^2 - 9x + 20$ (b) $2x^2 + 7x + 3$

3/3

(a) $x^2 - 9x + 20$: Split as $x^2 - 4x - 5x + 20 = x(x-4) - 5(x-4) = (x-5)(x-4)$. ✓ Correct. (1.5/1.5)

(b) $2x^2 + 7x + 3$: Split as $2x^2 + 6x + x + 3 = 2x(x+3) + 1(x+3) = (2x+1)(x+3)$. ✓ Correct. (1.5/1.5)

Q.2(vi) main — HCF of $12x^2y^3$ and $18x^3y^2$

3/3

$12 = 2^2 \times 3$, $18 = 2 \times 3^2$; x^2 and x^3 (min: x^2); y^3 and y^2 (min: y^2). HCF = $6x^2y^2$. ✓ *Correct. (3/3)*

Seemab showed the factorization of both numbers and identified shared factors correctly. Clean working.

Q.2(vii) main — Solve: $(2x)/3 + 1 = x/2 + 2$

3/3

LCD = 6: $4x + 6 = 3x + 12 \rightarrow x = 6$. ✓ *Correct. (3/3)*

Seemab showed clear step-by-step fraction clearing. Final answer $x = 6$ is correct.

Q.2(viii) main — Solve: $8x + 9 < 6x - 7$

3/3

$8x - 6x < -7 - 9 \rightarrow 2x < -16 \rightarrow x < -8$. Solution set: $\{x \mid x \in \mathbb{R}, x < -8\}$. ✓ *Correct. (3/3)*

Seemab wrote the solution set in correct set-builder notation. Clean and complete.

Q.2(ix) main — Distance between A(−2, 1) and B(1, 5)

3/3

$d = \sqrt{[(1-(-2))^2 + (5-1)^2]} = \sqrt{[3^2 + 4^2]} = \sqrt{[9+16]} = \sqrt{25} = 5$ units. ✓ *Correct. (3/3)*

Formula stated and correctly applied with arithmetic shown at each step.

Q.2(x) main — Midpoint of (2, 5) and (6, 9); state the quadrant

2/3

Seemab wrote: midpoint = $((6+2)/2, (9+5)/2) = (8/2, 14/2) = (4, 7)$. ✓ *Midpoint correct. (2/2)*

Quadrant: the question explicitly asks "state the quadrant in which it lies." No quadrant statement was written.

−1 mark: *quadrant not stated. Both coordinates positive → Quadrant I. This is a required part of the answer.*

Q.2(xi) main — Define: (a) chord (b) tangent to a circle (c) arc

3/3

Chord: "A line segment whose two endpoints both lie on the circle." ✓ *(1/1)*

Tangent: "A straight line touches the circle at exactly one point (the point of contact)." ✓ *(1/1)*

Arc: "The continuous part of the circumference (curved boundary) of the circle." ✓ *(1/1)*

All three definitions match the answer key language precisely. Excellent.

Section B Total	Attempted	Marks Earned	Marks Lost
11 questions × 3 marks	11/11	32/33	1 (Q2(x) quadrant omitted)

SECTION C — Long Questions (4 × 5 = 20 Marks)

Seemab answered the main question for all four long questions (Q3(i), Q3(ii), Q3(iii), Q3(iv) main).

Q.3(i) main — (a) Factorize $(x+1)(x+2)(x+3)(x+4)-24$ [3] (b) Square root of $4x^2+12xy+9y^2$ [2]

5/5

(a) Substitution factorization:

Seemab re-arranged as $(x+2)(x+3)(x+1)(x+4) - 24$, then grouped: $(x^2+5x+6)(x^2+5x+4) - 24$. Let $a = x^2+5x$, then $(a+6)(a+4) - 24 = a^2+10a+24-24 = a^2+10a = a(a+10) = (x^2+5x)(x^2+5x+10) = x(x+5)(x^2+5x+10)$. ✓ *Fully correct. (3/3)*

(b) Square root of $4x^2+12xy+9y^2$:

Seemab wrote $(2x)^2 + 2(2x)(3y) + (3y)^2 = \pm(2x+3y)$. ✓ *Correct recognition of perfect square and extraction. (2/2)*

Q.3(ii) main — (a) Three laws of logarithms [2] (b) Evaluate $\log 42$ and $\log(9/2)$ [3]

5/5

(a) Three laws:

1) $\log mn = \log m + \log n$ 2) $\log(m/n) = \log m - \log n$ 3) $\log m^n = n \log m$. ✓ *All three stated correctly. (2/2)*

(b) Evaluate: The exam paper provides $\log 2 = 0.3010$, $\log 3 = 0.4771$, $\log 7 = 0.8451$ and asks for (i) $\log 42$ and (ii) $\log(9/2)$.

(i) $\log 42$: $\log(2 \times 3 \times 7) = \log 2 + \log 3 + \log 7 = 0.3010 + 0.4771 + 0.8451 = 1.6232$. ✓ *Correct. (1.5/1.5)*

(ii) $\log(9/2)$: $\log 3^2/2 = 2 \log 3 - \log 2 = 2(0.4771) - 0.3010 = 0.9542 - 0.3010 = 0.6532$. ✓ *Correct. (1.5/1.5)*

Q.3(iii) main — (a) Solve $(x-2)/3 + (x+1)/2 = 4$ and verify [3] (b) Solve $3x-5 \leq 7$, $x \in \mathbb{R}$ with number line [2]

4/5

(a) Linear equation:

Seemab took LCD = 6: $2(x-2) + 3(x+1) = 24 \rightarrow 2x-4+3x+3 = 24 \rightarrow 5x-1 = 24$, then wrote "cross multiplication." She stopped at this point without completing the final step ($5x = 25$, $x = 5$) and without writing any verification.

–1 mark: solution left unfinished; $x = 5$ was never stated and no verification of the answer was written. The question explicitly says "solve and verify." Method and setup correctly established (2/3); the final line and the verification are missing (lost 1).

(b) Inequality: $3x \leq 7+5 = 12 \rightarrow x \leq 4$. Solution set: $\{x \mid x \in \mathbb{R}, x \leq 4\}$. Number line drawn with correct endpoint and shading. ✓ Correct and complete including number line. (2/2)

Q.3(iv) main — (a) Show A(–4,2), B(1,4), C(3,–1), D(–2,–3) form a square [4] (b) State distance formula [1]

3/5

(a) Square proof:

Seemab calculated all four side lengths correctly:

$AB = \sqrt{29}$, $BC = \sqrt{29}$, $CD = \sqrt{29}$, $DA = \sqrt{29}$. All sides equal. ✓ (2/2 for side calculations)

The question requires proving a square. FBISE marking requires either (i) proving diagonals equal ($AC = BD = \sqrt{58}$) OR (ii) proving adjacent sides perpendicular. Seemab stopped after showing all four sides equal — that proves a rhombus, not necessarily a square.

–2 marks: diagonal check entirely absent. Four equal sides prove a rhombus; the square proof additionally requires equal diagonals ($AC = BD = \sqrt{58}$) or a right-angle check. Neither is present. The answer key states: "The diagonals: $AC = \sqrt{58}$ and $BD = \sqrt{58}$. Since all sides are equal and the diagonals are equal, ABCD is a square." Both diagonal calculations must appear for the remaining 2 marks.

(b) Distance formula: Seemab stated $d = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$ correctly. ✓ (1/1)

Section C Total	Available	Marks Earned	Marks Lost
4 questions × 5 marks	20	17/20	3 (Q3(iii)(a) unfinished –1; Q3(iv)(a) diagonals –2)

Final Score Breakdown

Section	Available	Earned	%	Notes
Section A — MCQ	12	12	100%	All 12 correct — 7th consecutive perfect MCQ
Section B — Short Questions	33	32	97.0%	Lost 1 mark: Q2(x) quadrant not stated
Section C — Long Questions	20	17	85.0%	Lost 1 mark: Q3(iii)(a) solution unfinished; Lost 2 marks: Q3(iv)(a) diagonal proof absent
TOTAL	65	61	93.8%	Grade A+

Deduction Log (all 4 marks lost)

Question	Sub-part	Issue	Marks Lost
Q.2(x)	Quadrant statement	Midpoint (4,7) calculated correctly but no statement that both coordinates are positive → Quadrant I. The question explicitly requests it.	–1
Q.3(iii)(a)	Unfinished solution	Seemab set up the equation correctly through to $5x-1 = 24$ and wrote "cross multiplication," but stopped before completing the solution. The final step ($x = 5$) was never written, and no verification was attempted. Method earned 2; the completion and verification together cost 1.	–1
Q.3(iv)(a)	Diagonal check	All four side lengths computed correctly ($\sqrt{29}$ each) proving a rhombus, but square proof requires equal diagonals ($AC = BD = \sqrt{58}$) or a right-angle check. Calculation and conclusion both absent.	–2

Strengths in this exam

- Perfect MCQ (12/12) — all six topic areas covered cleanly
- Logarithm laws stated and applied flawlessly (Q3(ii) perfect 5/5)
- $\log 42 = \log 2 + \log 3 + \log 7 = 1.6232$ and $\log(9/2) = 0.6532$ both fully correct
- Factorization of $(x+1)(x+2)(x+3)(x+4)-24$ by substitution: correctly paired, substituted, and factored (Q3(i) perfect 5/5)
- Scientific notation, rational/irrational classification, and log expansion all clean in Section B
- Distance and midpoint formulas applied with no arithmetic errors
- Inequality $3x-5 \leq 7$ solved with correct set-builder notation and number line
- HCF by prime factorization: $6x^2y^2$ correctly identified with full factor trees shown
- Practical geometry definitions (chord, tangent, arc) match key language exactly

Errors in this exam

- **Q.2(x):** Quadrant statement omitted after correct midpoint. Easy mark left on the table — one sentence would have secured it.
- **Q.3(iii)(a):** Solution left unfinished. Setup and algebraic manipulation through $5x-1 = 24$ were correct, but the equation was not completed to $x = 5$, and the "verify" instruction was not acted on at all. Two marks were secured for the method; one mark was lost for leaving the solution mid-step.
- **Q.3(iv)(a):** Square proof incomplete. Four equal sides prove a rhombus. Two diagonal calculations ($AC = \sqrt{58}$, $BD = \sqrt{58}$) are required to confirm a square and are entirely absent. This is the largest single deduction on the paper (2 marks).

Revision targets from this exam

- **Square proof workflow (priority):** Four equal sides = rhombus. To confirm a square, always compute both diagonals and show $AC = BD$. Drill: $A(-1,0)$, $B(0,1)$, $C(1,0)$, $D(0,-1)$ — practise the full 6-calculation proof (4 sides + 2 diagonals) until automatic.
- **Finish what you start:** The equation $5x-1 = 24$ is one line from the answer. Forming the equation is not enough; always write the final value and any requested verification before moving on.
- **Re-read the instruction before moving on:** "Solve and verify" = two distinct deliverables. "State the quadrant" = a required sentence. Tick each instruction keyword explicitly during the exam.
- **Continuing pattern — stopping before all required sub-parts:** This is the 5th consecutive exam with this pattern. Q.2(x) quadrant and Q.3(iii)(a) both stopped one step before completion. Budget 30 seconds per answer to re-read the full question text.

Comparison with Prior Math Attempts

#	Date	Exam	Score	%	MCQ	Sec B	Sec C	Grade
1	18 Apr 2026	exam-01-ch1-2	65/65	100%	12/12	33/33	20/20	A+
2	22 Apr 2026	exam-03-ch4-5-7	60.5/65	93.1%	12/12	28.5/33	20/20	A+
3	25 May 2026	exam-04-ch4	57/65	87.7%	12/12	26/33	19/20	A
4	5 Jun 2026	exam-07-ch8	64/65	98.5%	11/12	33/33	20/20	A+
5	26 Jun 2026	exam-10-ch1-2-4-5-7-10	61/65	93.8%	12/12	32/33	17/20	A+

Section C 17/20 (85%) is Seemab's lowest Math Section C across all five attempts (prior were 20, 20, 19, 20), driven by the unfinished equation and the absent diagonal check — both are procedural gaps rather than conceptual ones. Section B remains very strong at 32/33 (97%). MCQ is back to 12/12 after the single error on exam-07.